

Student ID : _____

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1. Observed data $\{X[0], X[1], \dots, X[N-1]\}$ are mutually independent and normally distributed as $\sim N(0, \sigma^2)$. We estimate the unknown variance σ^2 as

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=0}^{N-1} X^2[n].$$

Judge whether this estimator is unbiased or not. Furthermore, compute the variance of $\hat{\sigma}^2$ to study its property in the limit of $N \rightarrow \infty$.

[Hint: $E[X^4[n]] = 3\sigma^4$.]

2. Observed data $\{X[0], X[1], \dots, X[N-1]\}$ are mutually independent and uniformly distributed as $\sim U(0, \theta)$, where $0 < \theta < \infty$. Construct an unbiased estimator for the unknown parameter θ and show that it is unbiased.

[Hint: $E[X[n]] = \frac{\theta}{2}$]

3. Consider the following observation

$$X[n] = Ar^n + W[n] \quad (n = 0, 1, \dots, N-1),$$

where $\{W[n]\}$ are mutually independent and normally distributed as $N(0, \sigma^2)$. r is a known constant. To estimate the unknown parameter A , compute the Cramér-Rao lower bound (CRLB) and derive the efficient estimator for A .

4. Observed data $\{X[0], X[1], \dots, X[N-1]\}$ are mutually independent and exponentially distributed as $E_x(\lambda)$. Obtain the maximum likelihood estimator of the unknown parameter λ . [Note that the exponential distribution has the probability density function of $p(x; \lambda) = \lambda \exp(-\lambda x)$ ($x \geq 0$); 0 ($x < 0$).]

5. X is a random variable uniformly distributed as $X \sim U\left[-\frac{1}{2}, \frac{1}{2}\right]$. Let us estimate a variable $\theta = \cos 2\pi X$ by using the following function

$$\hat{\theta} = aX^2 + b.$$

- (1) Compute the expectations for $E[X^2], E[X^4], E[\theta], E[\theta^2], E[\theta X^2]$.
- (2) Derive the parameter values of a, b based on the linear minimum mean square error (LMMSE) estimation.
- (3) Compute the minimum mean square error $Bmse$ corresponding to the parameter values a, b obtained in (2).

6. We predict λ -period future state $S(t + \lambda)$ of a stationary process by using its current state

$S(t)$ and T -period previous state $S(t - T)$ as

$$\hat{S}(t + \lambda) = aS(t) + bS(t - T).$$

Derive the optimal parameter values for a and b . Note that $R_{SS}(\tau) = \exp\left(-\frac{|\tau|}{T}\right)$, $0 < \lambda < T$.

Note:

- Please provide your answer in **clear handwriting**
- Dead-line for submission: **3 June 2026**